

PayShield – Complete Prototype Code

This document contains the complete three-file prototype: index.html, style.css, and script.js.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>PayShield</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <aside class="sidebar">
    <div class="brand">PayShield</div>
    <nav>
      <button class="nav-link active" data-view="verify">Verify Payment</button>
      <button class="nav-link" data-view="ledger">Ledger</button>
      <button class="nav-link" data-view="dashboard">Dashboard</button>
      <button class="nav-link" data-view="about">About</button>
    </nav>
  </aside>

  <main class="main-content">
    <section id="verify" class="view active">
      <div class="page-header">
        <h1>Payment Verification</h1>
        <p>Verify UPI payments using live requests or screenshot analysis.</p>
      </div>

      <div class="grid-2">
        <div class="panel">
          <h2>Layer 1 – Live Payment Request</h2>
          <p>Create a payment request and simulate buyer approval.</p>

          <form id="paymentForm">
            <label>Amount</label>
            <input id="paymentAmount" type="number" min="1" required>
```

```

<label>Customer Name</label>
<input id="customerName" type="text" required>

<button class="primary-btn" type="submit">Generate Request</button>
</form>

<div id="requestResult" class="hidden">
  <div class="qr-wrapper">
    <canvas id="qrCanvas" width="180" height="180"></canvas>
  </div>
  <p><strong>Transaction ID:</strong> <span id="transactionId"></span></p>
  <p><strong>Amount:</strong> ₹<span id="requestAmount"></span></p>
  <button id="approvePayment" class="success-btn">Simulate Buyer
Approval</button>
</div>
</div>

<div class="panel">
  <h2>Layer 2 — Screenshot Verification</h2>
  <p>Upload a payment screenshot and cross-check its details.</p>

  <div id="dropZone" class="drop-zone">
    <input id="screenshotInput" type="file" accept="image/*" hidden>
    <p>Drag & drop screenshot here</p>
    <button id="chooseFile" class="secondary-btn" type="button">Choose
Screenshot</button>
  </div>

  <img id="previewImage" class="preview hidden" alt="Payment screenshot
preview">

  <div id="analysisChecks" class="checks"></div>

  <label>Amount</label>
  <input id="verifyAmount" type="number" min="0" placeholder="Enter
amount">

  <label>Transaction / Reference ID</label>
  <input id="verifyTxnId" type="text" placeholder="Enter transaction ID">

  <button id="verifyBtn" class="primary-btn">Verify Payment</button>

  <div id="verdict" class="verdict hidden"></div>
</div>

```

```
</div>
</section>
```

```
<section id="ledger" class="view">
  <div class="page-header">
    <h1>Bank & UPI Ledger</h1>
    <p>Confirmed transactions stored in the local ledger.</p>
  </div>
  <div class="panel">
    <div class="table-wrap">
      <table>
        <thead>
          <tr>
            <th>Transaction ID</th>
            <th>Customer</th>
            <th>Amount</th>
            <th>Status</th>
            <th>Date</th>
          </tr>
        </thead>
        <tbody id="ledgerBody"></tbody>
      </table>
    </div>
  </div>
</section>
```

```
<section id="dashboard" class="view">
  <div class="page-header">
    <h1>Vendor Dashboard</h1>
    <p>Overview of verified payment activity.</p>
  </div>
```

```
<div class="stats">
  <div class="stat-card">
    <span>Total Transactions</span>
    <strong id="totalTransactions">0</strong>
  </div>
  <div class="stat-card">
    <span>Total Amount</span>
    <strong>₹<span id="totalAmount">0</span></strong>
  </div>
  <div class="stat-card">
    <span>Verified Scans</span>
    <strong id="verifiedScans">0</strong>
  </div>
```

```
<div class="stat-card">
  <span>Suspicious / Fake</span>
  <strong id="flaggedScans">0</strong>
</div>
</div>
```

```
<div class="panel">
  <h2>Recent Scan History</h2>
  <div id="historyList"></div>
  <button id="clearHistory" class="danger-btn">Clear Scan History</button>
</div>
</section>
```

```
<section id="about" class="view">
  <div class="page-header">
    <h1>About PayShield</h1>
    <p>A prototype for helping vendors verify UPI payment claims.</p>
  </div>
```

```
<div class="about-grid">
  <div class="panel">
    <h2>What it does</h2>
    <p>
      PayShield provides two verification layers: a simulated live payment
      request and screenshot-based verification.
    </p>
  </div>
```

```
<div class="panel">
  <h2>Technology</h2>
  <p>
    HTML, CSS, JavaScript, Canvas API, FileReader API, DOM manipulation,
    and localStorage.
  </p>
</div>
```

```
<div class="panel">
  <h2>Prototype limitation</h2>
  <p>
    The QR shown here is illustrative and not a real scannable UPI QR.
    Full OCR, trained ML classification, backend integration, and real
    bank/UPI APIs are not implemented in this prototype.
  </p>
</div>
</div>
```

```
</section>
</main>
```

```
<div id="toast" class="toast"></div>
```

```
<script src="script.js"></script>
</body>
</html>
```

style.css

```
* {
  box-sizing: border-box;
}
```

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background: #07111f;
  color: #e8eef7;
  min-height: 100vh;
}
```

```
.sidebar {
  position: fixed;
  left: 0;
  top: 0;
  bottom: 0;
  width: 240px;
  padding: 28px 18px;
  background: #0b1728;
  border-right: 1px solid #1c2d42;
}
```

```
.brand {
  font-size: 26px;
  font-weight: 800;
  margin-bottom: 35px;
}
```

```
nav {
  display: flex;
  flex-direction: column;
  gap: 10px;
```

```
}  
  
.nav-link {  
  border: 0;  
  background: transparent;  
  color: #aab8ca;  
  padding: 13px 15px;  
  text-align: left;  
  border-radius: 10px;  
  cursor: pointer;  
  font-size: 15px;  
}
```

```
.nav-link:hover,  
.nav-link.active {  
  background: #16263b;  
  color: #ffffff;  
}
```

```
.main-content {  
  margin-left: 240px;  
  padding: 35px;  
}
```

```
.view {  
  display: none;  
}
```

```
.view.active {  
  display: block;  
}
```

```
.page-header {  
  margin-bottom: 25px;  
}
```

```
.page-header h1 {  
  margin-bottom: 8px;  
}
```

```
.page-header p,  
.panel p {  
  color: #aab8ca;  
  line-height: 1.6;  
}
```

```
.grid-2 {  
  display: grid;  
  grid-template-columns: repeat(2, minmax(0, 1fr));  
  gap: 22px;  
}
```

```
.panel {  
  background: #0d1b2d;  
  border: 1px solid #1c3047;  
  border-radius: 16px;  
  padding: 24px;  
  margin-bottom: 22px;  
}
```

```
.panel h2 {  
  margin-top: 0;  
}
```

```
label {  
  display: block;  
  margin: 16px 0 7px;  
  font-size: 14px;  
  color: #b9c7d8;  
}
```

```
input {  
  width: 100%;  
  padding: 12px;  
  border-radius: 9px;  
  border: 1px solid #30455d;  
  background: #081522;  
  color: #ffffff;  
  outline: none;  
}
```

```
button {  
  font: inherit;  
}
```

```
.primary-btn,  
.secondary-btn,  
.success-btn,  
.danger-btn {  
  border: 0;
```

```
padding: 12px 17px;
border-radius: 9px;
cursor: pointer;
margin-top: 18px;
}
```

```
.primary-btn {
  background: #d7ad45;
  color: #101010;
  font-weight: 700;
}
```

```
.secondary-btn {
  background: #20354d;
  color: #ffffff;
}
```

```
.success-btn {
  background: #1e9b70;
  color: #ffffff;
}
```

```
.danger-btn {
  background: #8e3140;
  color: #ffffff;
}
```

```
.drop-zone {
  border: 2px dashed #39536f;
  border-radius: 13px;
  padding: 30px;
  text-align: center;
  margin: 18px 0;
}
```

```
.preview {
  display: block;
  max-width: 100%;
  max-height: 300px;
  margin: 18px auto;
  border-radius: 10px;
}
```

```
.qr-wrapper {
  display: flex;
```

```
justify-content: center;
margin: 20px 0;
}
```

```
#qrCanvas {
  background: white;
  border-radius: 8px;
}
```

```
.check {
  padding: 10px 12px;
  border-radius: 8px;
  background: #12263b;
  margin: 8px 0;
}
```

```
.verdict {
  padding: 18px;
  border-radius: 10px;
  margin-top: 18px;
  font-weight: 700;
}
```

```
.verdict.genuine {
  background: #123d31;
  color: #8ce8c5;
}
```

```
.verdict.suspicious {
  background: #4a3915;
  color: #f4d77e;
}
```

```
.verdict.fake {
  background: #461d27;
  color: #ff9eae;
}
```

```
.table-wrap {
  overflow-x: auto;
}
```

```
table {
  width: 100%;
  border-collapse: collapse;
```

```
}  
}
```

```
th,  
td {  
  padding: 13px;  
  border-bottom: 1px solid #23364b;  
  text-align: left;  
}
```

```
th {  
  color: #c8d4e2;  
}
```

```
.stats {  
  display: grid;  
  grid-template-columns: repeat(4, 1fr);  
  gap: 18px;  
  margin-bottom: 24px;  
}
```

```
.stat-card {  
  background: #0d1b2d;  
  border: 1px solid #1c3047;  
  border-radius: 15px;  
  padding: 22px;  
}
```

```
.stat-card span {  
  color: #9eafc2;  
  display: block;  
  margin-bottom: 12px;  
}
```

```
.stat-card strong {  
  font-size: 28px;  
}
```

```
.about-grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 20px;  
}
```

```
.history-item {  
  border-bottom: 1px solid #23364b;  
}
```

```
padding: 14px 0;  
}
```

```
.hidden {  
  display: none !important;  
}
```

```
.toast {  
  position: fixed;  
  right: 25px;  
  bottom: 25px;  
  background: #172b42;  
  color: white;  
  padding: 13px 18px;  
  border-radius: 9px;  
  opacity: 0;  
  transform: translateY(15px);  
  pointer-events: none;  
  transition: 0.25s;  
}
```

```
.toast.show {  
  opacity: 1;  
  transform: translateY(0);  
}
```

```
@media (max-width: 900px) {  
  .grid-2,  
  .about-grid,  
  .stats {  
    grid-template-columns: 1fr 1fr;  
  }  
}
```

```
@media (max-width: 650px) {  
  .sidebar {  
    position: static;  
    width: 100%;  
    height: auto;  
    border-right: 0;  
    border-bottom: 1px solid #1c2d42;  
  }  
}
```

```
nav {  
  flex-direction: row;
```

```
    overflow-x: auto;
  }

  .nav-link {
    white-space: nowrap;
  }

  .main-content {
    margin-left: 0;
    padding: 20px;
  }

  .grid-2,
  .about-grid,
  .stats {
    grid-template-columns: 1fr;
  }
}
```

script.js

```
const LEDGER_KEY = "payshield_ledger";
const HISTORY_KEY = "payshield_history";

const $ = (id) => document.getElementById(id);

function getLedger() {
  return JSON.parse(localStorage.getItem(LEDGER_KEY) || "[]");
}

function saveLedger(data) {
  localStorage.setItem(LEDGER_KEY, JSON.stringify(data));
}

function getHistory() {
  return JSON.parse(localStorage.getItem(HISTORY_KEY) || "[]");
}

function saveHistory(data) {
  localStorage.setItem(HISTORY_KEY, JSON.stringify(data));
}

function showToast(message) {
  const toast = $("toast");
```

```
toast.textContent = message;
toast.classList.add("show");

setTimeout(() => {
  toast.classList.remove("show");
}, 2500);
}

function seedLedger() {
  if (getLedger().length > 0) return;

  const seed = [
    {
      id: "PS100001",
      customer: "Demo Customer",
      amount: 500,
      status: "CONFIRMED",
      date: new Date().toLocaleString()
    },
    {
      id: "PS100002",
      customer: "Test User",
      amount: 1200,
      status: "CONFIRMED",
      date: new Date().toLocaleString()
    }
  ];

  saveLedger(seed);
}

function navigate(viewName) {
  document.querySelectorAll(".view").forEach(view => {
    view.classList.remove("active");
  });

  document.querySelectorAll(".nav-link").forEach(btn => {
    btn.classList.remove("active");
  });

  $(viewName).classList.add("active");

  const button = document.querySelector(`[data-view="${viewName}"]`);
  if (button) button.classList.add("active");
}
```

```

    if (viewName === "ledger") renderLedger();
    if (viewName === "dashboard") renderDashboard();
  }

  document.querySelectorAll(".nav-link").forEach(button => {
    button.addEventListener("click", () => {
      navigate(button.dataset.view);
    });
  });

  let currentRequest = null;

  $("paymentForm").addEventListener("submit", (event) => {
    event.preventDefault();

    const amount = Number($("#paymentAmount").value);
    const customer = $("#customerName").value.trim();

    if (!amount || amount <= 0 || !customer) {
      showToast("Please enter valid payment details.");
      return;
    }

    const transactionId =
      "PS" + Date.now().toString().slice(-8);

    currentRequest = {
      id: transactionId,
      customer,
      amount,
      status: "PENDING",
      date: new Date().toLocaleString()
    };

    $("#transactionId").textContent = transactionId;
    $("#requestAmount").textContent = amount.toFixed(2);
    $("#requestResult").classList.remove("hidden");

    drawMockQR(transactionId, amount);
    showToast("Payment request generated.");
  });

  function drawMockQR(transactionId, amount) {
    const canvas = $("#qrCanvas");
    const ctx = canvas.getContext("2d");

```

```

ctx.fillStyle = "#ffffff";
ctx.fillRect(0, 0, canvas.width, canvas.height);

const size = 9;
const seed = transactionId.length + Math.round(amount);

for (let y = 0; y < canvas.height / size; y++) {
  for (let x = 0; x < canvas.width / size; x++) {
    const value = (x * 17 + y * 31 + seed) % 7;

    if (value < 3) {
      ctx.fillStyle = "#000000";
      ctx.fillRect(x * size, y * size, size, size);
    }
  }
}

drawFinder(ctx, 10, 10, size);
drawFinder(ctx, canvas.width - 70, 10, size);
drawFinder(ctx, 10, canvas.height - 70, size);
}

function drawFinder(ctx, startX, startY, size) {
  ctx.fillStyle = "#000000";
  ctx.fillRect(startX, startY, size * 7, size * 7);

  ctx.fillStyle = "#ffffff";
  ctx.fillRect(startX + size, startY + size, size * 5, size * 5);

  ctx.fillStyle = "#000000";
  ctx.fillRect(startX + size * 2, startY + size * 2, size * 3, size * 3);
}

$("approvePayment").addEventListener("click", () => {
  if (!currentRequest) return;

  const ledger = getLedger();

  const confirmed = {
    ...currentRequest,
    status: "CONFIRMED"
  };

  ledger.push(confirmed);

```

```
saveLedger(ledger);

currentRequest = null;
$("#requestResult").classList.add("hidden");
$("#paymentForm").reset();

showToast("Payment confirmed and added to ledger.");
renderLedger();
});

const dropZone = $("#dropZone");
const screenshotInput = $("#screenshotInput");

$("#chooseFile").addEventListener("click", () => {
  screenshotInput.click();
});

dropZone.addEventListener("dragover", (event) => {
  event.preventDefault();
  dropZone.classList.add("dragging");
});

dropZone.addEventListener("dragleave", () => {
  dropZone.classList.remove("dragging");
});

dropZone.addEventListener("drop", (event) => {
  event.preventDefault();
  dropZone.classList.remove("dragging");

  const file = event.dataTransfer.files[0];
  if (file) handleScreenshot(file);
});

screenshotInput.addEventListener("change", (event) => {
  const file = event.target.files[0];
  if (file) handleScreenshot(file);
});

let imageAnalysis = null;

function handleScreenshot(file) {
  if (!file.type.startsWith("image/")) {
    showToast("Please upload an image file.");
    return;
  }
}
```

```
}
```

```
const reader = new FileReader();
```

```
reader.onload = (event) => {  
  $("previewImage").src = event.target.result;  
  $("previewImage").classList.remove("hidden");
```

```
  analyzelImage(event.target.result);  
};
```

```
reader.readAsDataURL(file);  
}
```

```
function analyzelImage(dataUrl) {  
  const image = new Image();
```

```
  image.onload = () => {  
    const canvas = document.createElement("canvas");  
    const maxWidth = 500;
```

```
    const scale = Math.min(1, maxWidth / image.width);
```

```
    canvas.width = Math.round(image.width * scale);  
    canvas.height = Math.round(image.height * scale);
```

```
    const ctx = canvas.getContext("2d");  
    ctx.drawImage(image, 0, 0, canvas.width, canvas.height);
```

```
    const pixels = ctx.getImageData(  
      0,  
      0,  
      canvas.width,  
      canvas.height  
    ).data;
```

```
    let greenPixels = 0;  
    let brightPixels = 0;
```

```
    for (let i = 0; i < pixels.length; i += 4) {  
      const r = pixels[i];  
      const g = pixels[i + 1];  
      const b = pixels[i + 2];
```

```
      if (g > r * 1.2 && g > b * 1.1 && g > 80) {
```

```
        greenPixels++;
    }

    if (r > 220 && g > 220 && b > 220) {
        brightPixels++;
    }
}

const totalPixels = pixels.length / 4;
const greenRatio = greenPixels / totalPixels;
const brightRatio = brightPixels / totalPixels;

const ratio = image.width / image.height;

imageAnalysis = {
    width: image.width,
    height: image.height,
    ratio,
    greenRatio,
    brightRatio
};

renderChecks();
showToast("Screenshot analyzed.");
};

image.src = dataUrl;
}

function renderChecks() {
    if (!imageAnalysis) return;

    const checks = [];

    const ratioPass =
        imageAnalysis.ratio > 0.45 &&
        imageAnalysis.ratio < 1.2;

    const greenPass =
        imageAnalysis.greenRatio > 0.005;

    const surfacePass =
        imageAnalysis.brightRatio > 0.10;

    checks.push({
```

```

    label: "Screen / aspect ratio",
    pass: ratioPass
  });

  checks.push({
    label: "Success indicator / green signal",
    pass: greenPass
  });

  checks.push({
    label: "Surface consistency",
    pass: surfacePass
  });

  $("analysisChecks").innerHTML = checks
    .map(check => `
      <div class="check">
        ${check.pass ? "✓" : "✗"} ${check.label}
      </div>
    `)
    .join("");
}

$("verifyBtn").addEventListener("click", () => {
  const amount = Number($("verifyAmount").value);
  const txnId = $("verifyTxnId").value.trim();

  if (!imageAnalysis) {
    showToast("Please upload a screenshot first.");
    return;
  }

  if (!amount || !txnId) {
    showToast("Enter amount and transaction ID.");
    return;
  }

  const ledger = getLedger();

  const matchingTransaction = ledger.find(
    transaction =>
      transaction.id === txnId &&
      Number(transaction.amount) === amount &&
      transaction.status === "CONFIRMED"
  );

```

```
const checks = document.querySelectorAll(".check");
const passedChecks = [...checks].filter(
  check => check.textContent.trim().startsWith("✓")
).length;
```

```
let verdict;
let className;
```

```
if (matchingTransaction && passedChecks >= 2) {
  verdict = "GENUINE";
  className = "genuine";
} else if (passedChecks >= 2) {
  verdict = "SUSPICIOUS";
  className = "suspicious";
} else {
  verdict = "FAKE";
  className = "fake";
}
```

```
const verdictBox = $("verdict");
```

```
verdictBox.className = `verdict ${className}`;
verdictBox.textContent =
  `Verdict: ${verdict}`;
```

```
verdictBox.classList.remove("hidden");
```

```
const history = getHistory();
```

```
history.unshift({
  amount,
  txnId,
  verdict,
  date: new Date().toLocaleString()
});
```

```
saveHistory(history.slice(0, 50));
```

```
renderDashboard();
});
```

```
function renderLedger() {
  const ledger = getLedger();
```

```

$("ledgerBody").innerHTML = ledger
  .slice()
  .reverse()
  .map(transaction => `
    <tr>
      <td>${escapeHtml(transaction.id)}</td>
      <td>${escapeHtml(transaction.customer)}</td>
      <td>₹${Number(transaction.amount).toFixed(2)}</td>
      <td>${escapeHtml(transaction.status)}</td>
      <td>${escapeHtml(transaction.date)}</td>
    </tr>
  `)
  .join("");
}

function renderDashboard() {
  const ledger = getLedger();
  const history = getHistory();

  const totalAmount = ledger.reduce(
    (sum, transaction) => sum + Number(transaction.amount),
    0
  );

  const verified = history.filter(
    item => item.verdict === "GENUINE"
  ).length;

  const flagged = history.filter(
    item => item.verdict === "SUSPICIOUS" ||
      item.verdict === "FAKE"
  ).length;

  $("totalTransactions").textContent = ledger.length;
  $("totalAmount").textContent = totalAmount.toFixed(2);
  $("verifiedScans").textContent = verified;
  $("flaggedScans").textContent = flagged;

  $("historyList").innerHTML = history.length
    ? history.slice(0, 10).map(item => `
      <div class="history-item">
        <strong>${escapeHtml(item.verdict)}</strong>
        <div>₹${Number(item.amount).toFixed(2)} – ${escapeHtml(item.txnId)}</
div>
        <small>${escapeHtml(item.date)}</small>

```

```
        </div>
      `).join("")
      : "<p>No scan history yet.</p>";
    }
  }

```

```
$("clearHistory").addEventListener("click", () => {
  saveHistory([]);
  renderDashboard();
  showToast("Scan history cleared.");
});

```

```
function escapeHtml(value) {
  return String(value)
    .replaceAll("&", "&amp;")
    .replaceAll("<", "&lt;")
    .replaceAll(">", "&gt;")
    .replaceAll('"', "&quot;")
    .replaceAll("'", "&#039;");
}

```

```
seedLedger();
renderLedger();
renderDashboard();

```